Implications of Tidal Heating on Titan's Subsurface Ocean Habitability

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Abstract

Titan's non-zero Love number ($k_2 = 0.616 \pm 0.067$) has long been regarded as evidence of a subsurface ocean, generated by tidal heating as Titan orbits Saturn. The availability of a global liquid layer would provide a key astrobiological environment for life in the Solar System, following the assumed requirements for prebiotic reactions to occur are met. However, Titan's capacity to host a subsurface ocean under Cassini observational constraints has recently come under question, with suggestion that the moon's measured energy dissipation is too large to indicate a liquid layer at depth. Consequently, understanding the control that Titan's tidal heating, and associated tidal parameters (k_2, Q), exert on observing interior structure and inferring ongoing dynamics is vital for establishing habitability. This study therefore investigated Titan's potential for life through examining its capacity to host an ocean, considering the influence of changing tidal parameters and energy dissipation over time. Through application of the PyALMA3 model, relevant k_2 and Q values were extracted for model Titan interiors that aligned with Cassini data, indicating that both a fully-ocean interior and solid-ice interior were possible within current boundary conditions. Subsequent analysis of interior energy dissipation indicated that a solid-ice Titan would support a subsurface ocean after 0.71 Gyrs of heating at its current orbital state, with its orbital migration of 11cm/yr having a negligible effect on tidal heating. Despite current debate regarding Titan's composition, it can therefore be suggested that tidal dissipation alone may produce a habitable ocean environment within the main-sequence lifetime of the Sun.

Contents

1 Introduction

1.1 Astrobiological Overview

Saturn's icy moon Titan has been highlighted as an object of astrobiological significance in the Solar System due to its unique structure and host of organic materials. It is the only satellite body with a dense atmosphere, and the only other site of surface liquid in the Solar System, presenting a diverse surface geology morphologically similar to Earth despite the differences in composition (Sohl *et al.*, 2014). Surface features include hydrocarbon lakes, key prebiotic compounds like hydrogen cyanide (Raulin, 2008), and candidate cryovolcanic regions, therefore providing an environment resembling that of primordial Earth. Titan's focus as an astrobiological target however is not limited to its surface, with observations by NASA's Cassini mission indicating that Titan hosts a global subsurface ocean (Iess *et al.*, 2012). As the minimum requirements for life include an abundant supply of liquid solvent, availability of organic materials, and an energy source to drive abiogenetic reactions (Maruyama *et al.*, 2019), determining the presence of a sustained global ocean would be crucial for evaluating Titan's habitability potential. The establishment of exchange processes between Titan's interior and surface, via impact cratering or cryovolcanism, could subsequently be a driver for global habitability by sustaining a supply of liquid, prebiotic material, and energy.

1.2 Titan's Tides

Planetary satellites experience orbit-induced tides which can be detected via measurements of their gravity field. Titan is tidally locked to Saturn and orbits with a non-negligible eccentricity (e = 0.028), resulting in periodic tidal stresses over its 16-day orbit and subsequent tidal dissipation. This eccentricity induces a short-term variation of the quadrupole tidal field proportional to $1/r^3$, where r is the distance between Titan and its orbital barycentre, resulting in changes to the moon's physical structure and gravity. The quantification of this tidal response comes from the degree-2 Love number k_2 that describes how deformable the body is to tides (Downey and Nimmo, 2025), with $k_2 = 0$ for a perfectly rigid body and $k_2 = 3/2$ for an incompressible liquid body. Analysis of Cassini's first six gravity flybys (Iess *et al.*, 2010; Iess *et al.*, 2012) through measurement of Cassini's orbital acceleration inferred that Titan's fluctuating deformability over its orbit was consistent with the presence of a subsurface ocean at depth. Subsequent analysis of radio-tracking data from all ten flybys derived estimates of $k_2 = 0.616 \pm 0.067$ (Durante *et al.*, 2019), and was accepted as an indicator of a liquid interior layer.

However, recent re-evaluation of Cassini observations has implied that the observed tidal energy dissipation is too high for there to be a subsurface ocean (Petricca *et al.*, 2025), as a liquid layer is expected to damp any tidal dissipation produced below it. This high tidal dissipation is associated with a low tidal quality factor Q (Q=5), where a small Q indicates a greater tidal lag and thus greater tidal dissipation (Downey and Nimmo, 2025). There also remains debate over the measurement of Titan's k_2 value, with a substantially lower Love number recently being stated $k_2 = 0.375 \pm 0.06$ (Goossens *et al.*, 2024) compared to previous studies (Iess *et al.*, 2010; Iess *et al.*, 2012; Durante *et al.*, 2019). Whilst a much smaller k_2 may indicate a lower-density, ammonia or water-rich ocean, it could also represent a less deformable body and preclude a subsurface liquid layer. The absence of a global ocean would overall limit Titan's habitability and restrict astrobiological investigations to the surface, where the surface-interior interactions thought to be crucial in supporting life may also be altered.

As k_2 is dependent on interior composition, measuring its change with respect to different Titan interior models can inform what range of k_2 would be most suitable for global habitability. The aim of this investigation is to therefore determine which tidal parameters would be the most habitable for Titan, in terms of supporting a subsurface ocean over a sustained period of time, allowing for necessary prebiotic reactions to take place. Additionally, as Titan is migrating rapidly outward from Saturn at a rate of 11cm/yr due to its resonance-locked orbit (Lainey *et al.*, 2020), it's important to consider the effect of migration on dampening energy dissipation and melting of the interior. With the upcoming NASA Dragonfly mission set to arrive at Titan in 2034 (NASA, 2023), this research may also provide necessary context for determining parameters for subsurface habitability.

2 Methods

This section will present the methods used to provide the interior modelling constraints (Section 2.1), the extraction of the tidal Love number k_2 using the PyALMA3 model (Section 2.2), and the application to energy dissipation within the interior (Section 2.3).

2.1 Interior Modelling

Titan's mass and bulk density measurements indicate a composition of rock and ice (Fortes, 2012), but specific density distributions are challenging to constrain due to large uncertainties in current observations. Different interior models needed to be tested to determine respective changes in k_2, the parameters of which were based on the methods applied by Petricca *et al.*, 2025; as a recent Titan study focused on subsurface ocean availability from tidal observations, the physical assumptions made for Titan's composition were directly relevant to this investigation. This includes accounting for a rocky core, a high-pressure (hp) ice (Ice VI, Ice V, Ice III) layer, an ocean layer, and a decoupled low-pressure ice (Ice *Ih*) shell, each assumed to be radial layers of uniform density. These models however do not consider differentiation of the ice shell into conductive and convective layers, which was found to have little contribution to the measured tidal dissipation (Petricca *et al.*, 2025). The ice shell (100km) and core (2000km) thickness were assumed to be uniform and constant, therefore isolating variations in k_2 to changes in hp-ice and ocean thickness within a 475km layer, as Titan is of fixed radius 2575km.

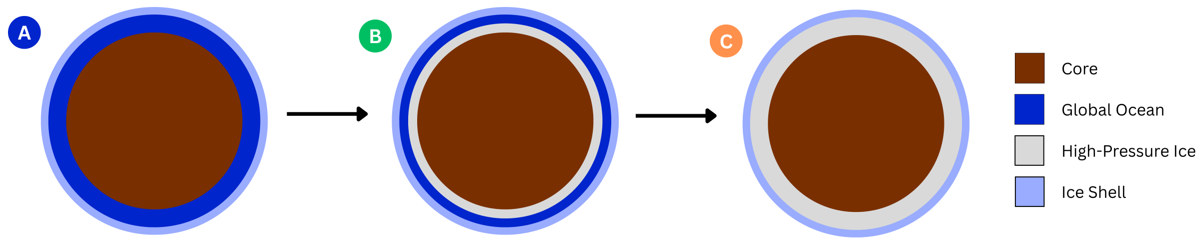

Figure 1: Illustrations of the 3 model interiors being considered in this investigation. From left to right there is the fully-liquid (Model A), hp-ice and ocean (Model B), and solid-ice (Model C). Generally, Titan has been considered to have a Model B interior, but recent Cassini re-evaluation has highlighted Model C as a more plausible composition (Petricca *et al.*, 2025).

The investigated models iterated from a 3 layer (core→ocean→shell) Model A with maximum ocean thickness (475km), to a 4 layer (core→hp-ice→ocean→shell) Model B of reducing ocean thickness, to a final 3 layer (core→ocean→shell) Model C with maximum hp-ice thickness. The full list of parameters used, including relevant rheological properties, can be found in Appendix Table 2. The visualisation of Models A, B and C are displayed in Figure 1.

2.2 Love Number k_2 Modelling

Petricca *et al.*, 2025 used the PyALMA3 software (Styczinski *et al.*, 2024), a Python package that calculates tidal Love numbers given specific interior constraints, to compute Titan's tidal response assuming a rocky core and hydrosphere. The same software was used in this investigation to determine the complex k_2 value of different interior compositions, and subsequently extract the tidal quality factor Q (Eqn.1).

$$Q = -\frac{Im(k_2)}{Re(k_2} \tag{1}$$

For clarity, whenever k_2 is mentioned without explicitly being referenced as imaginary Im(k_2), the Love number is assumed to be real Re(k_2). This includes the observed k_2 values from the Cassini data, as it is the measure of Titan's deformability that is being compared. The Im(k_2) instead measures the phase lag in the tidal response (Rappaport *et al.*, 2008). The set-up of PyALMA3 followed the example Julia code used to investigate the tidal deformation of Enceladus' interior (Styczinski *et al.*, 2024), inputting Titan's parameters (Appendix Table 2). Initial tests following the Julia code to examine changing shell thickness on k_2 are demonstrated in the Appendix Figures 8 - 11. To extract the complex k_2 Love number, a viscoelastic interior was assumed, applying a temperature-dependent viscosity profile in the interior according to the Arrhenius Law (Eqn.2). The temperature beneath the ice shell was also assumed to scale with radius (Eqn.3). Both equations were taken directly from the Julia code, where η_m is the ice-melt viscosity, E_a is the activation energy, R_g is the gas constant, r_t is the Titan radius, and r_b is the radius at the base of the *Ih* shell.

$$\eta = \eta_m \, exp(\frac{E_a}{R_g T_m} \, [\frac{T_m}{T} - 1)] \tag{2}$$

$$T(r) = {T_m}^{(r-r_t)/(r_b-r_t)} \; {T_s}^{(r_b-r)/(r_b-r_t)} \tag{3}$$

The temperature at the base of the *Ih* shell was assumed as its melting point ($T_m = 250K$) for equilibrium with an ocean, whilst surface temperature was constrained by Cassini observations ($T_s = 94K$) (Jennings *et al.*, 2019). Full parameters are again found in Appendix Table 2.

2.3 Energy Dissipation

This section details the modelling of energy dissipation using original Python scripts. To examine the impact of tidal heating on subsurface ocean habitability, only tidal forcing is considered as the driver of temperature evolution. Radiogenic heating is also estimated to be an order of magnitude smaller than Titan's tidal dissipation (3-4TW), and would therefore not significantly alter the estimated energy availability (Petricca *et al.*, 2025).

2.3.1 Rate of Tidal Dissipation

The simplified rate of energy dissipation within a synchronously rotating body in an eccentric orbit (Eqn.4), assumes that Titan is incompressible, that rotation is uniform and synchronous, and that orbital eccentricity is small (Wisdom, 2008). Titan has a small eccentricity (e = 0.028), and as the main aim of the investigation was to examine the effect of k_2 and Q on tidal dissipation, these assumptions were applied to identify general trends. The published k_2 values $k_2 = 0.616 \pm 067$ (Durante *et al.*, 2019) and $k_2 = 0.375 \pm 0.06$ (Goossens *et al.*, 2024) provided upper and lower k_2 limits, whilst the k_2/Q range of 0.058-0.12 (Downey and Nimmo, 2025) constrained the span of Q values, displayed in Table 1. This gave a baseline range to compare the extracted PyALMA3 values.

$$\frac{dE}{dt} = \frac{21}{2}\frac{k_2}{Q}\frac{GM^2 nR^5 e^2}{a^6} \tag{4}$$

Where G is the gravitational constant, M is the primary body's mass (Saturn), R is Titan's radius, and n is the orbital mean motion. Full parameters are found in Appendix Table 2.

k_2	k_2/Q	Q
$0.375 - 0.060$	0.058	54.3
$0.375 - 0.060$	0.12	5.69
$0.616 + 0.067$	0.058	118
$0.616 + 0.067$	0.12	2.62

Table 1: Upper and lower limits for k_2 and Q taken from Cassini analysis, calculating Q from k_2 and k_2/Q observations.

Using Eqn.4, a 2D-map of the tidal dissipation rate was modelled as a function of k_2 and Q from the published boundary values (Table 1). The corresponding heating rate was computed over a grid of k_2 and Q and stored as a 2D array, which was then visualised as a heat-map using a logarithmic colour scale. The trajectory of k_2 and Q under changing hp-ice layer thickness was then overlaid to demonstrate the path of the interior models through the parameter space. This provided a direct comparison between current observed tidal parameters and potential interior structures.

2.3.2 Heat Diffusion

To establish the melting timescales for the fully-solid interior (Model C), a 1D heat diffusion model was produced assuming spherical symmetry and uniform radial layers. As this model spans the hp-ice layer, the inner and outer radius was fixed at the core→hp-ice boundary (2000km) and the hp-ice→ice shell boundary (2475km). The corresponding energy dissipation rate (Eqn.4) was divided over the ice-layer volume to determine E_{tidal}, the volumetric heat source of tidal dissipation at the inner boundary. Although tidal dissipation is heterogeneously distributed with depth, assuming a uniform dissipation distribution throughout the ice layer for a first-order approximation of melting has previously been used in thermal evolution studies (Tobie *et al.*, 2005). The equation of heat diffusion in tidally-locked exoplanets (Murgia, 2022), was adapted to only consider radial components. The constant heat source was also assumed to be E_{tidal} rather than radiogenic decay (Eqn.5).

$$\frac{\delta T}{\delta t} = \frac{\alpha}{r^2}\frac{\delta}{\delta r}(r^2\frac{\delta T}{\delta r}) + \frac{E_{tidal}}{c_p \rho} \tag{5}$$

Where ρ is density, c_p is specific heat capacity, and α is thermal diffusivity of the hp-ice. The thermal properties of the hp-ice layer, its constituent pressure-dependent ice phases (Ice VI, Ice V, Ice III), and associated boundary conditions for modelling can be found in Appendix Table 3. The melting of the hp-ice needed to be assessed against a pressure-dependent melting curve to account for the layer's ice phases, therefore a hydrostatic pressure profile was computed (Eqn. 6) to determine pressure with increasing depth.

$$P(r) = P_{outer} + \rho g\ (R_{outer} - r) \tag{6}$$

Parameters from the Simon-Glatzel equation, the empirical formula describing how melting temperature changes with pressure, for Ice *Ih*, Ice III, and Ice V were applied from Choukroun and Grasset, 2010 (Appendix Table 3) to determine melting temperature at specific ice thickness. As the dominant ice-phase, Ice VI was modelled using the standard form of the Simon-Glatzel equation (Eqn.7) to provide the most physically accurate melting criterion across the layer, where a and c are empirical parameters derived from experimental data, P_m and T_m are melting pressure and temperature, and P_0 and T_0 are reference pressure and temperatures.

As minor phases, linear approximations were applied for Ice *Ih*, Ice III, and Ice V for simplicity. Melting was identified as the first time step where the interior temperature exceeded the specified ice melting temperature.

$$P_m = P_0 + a(\frac{T_m}{T_0})^c \ - 1 \tag{7}$$

The heat diffusion equation (Eqn.5) was discretised using a finite difference scheme with the RK45 solver from the Python SciPy package. This was applied to a uniform radial grid of 200 points and integrated over a 5 Gyr lifetime to correspond to the remaining main-sequence lifetime of the Sun, following which the evolution to the red-giant branch will have significant impact on the orbital evolution of Solar System planets (Esseldeurs *et al.*, 2026). To test whether Titan's orbital expansion of 11cm/yr had a significant reduction on energy dissipation over time, Titan's the semi-major axis was updated at each time-step, assuming a constant expansion rate. This was then compared to the heat diffusion at a constant semi-major axis over 5 Gyrs to establish the fractional change in energy dissipation.

3 Results

3.1 Complex Love Number k_2

The range of viscoelastic k_2 numbers extracted from the PyALMA3 software (Section 2.2) with changing hp-ice thickness are displayed in Figure 2, including both $\mathrm{Re}(k_2)$ and $\mathrm{Im}(k_2)$.

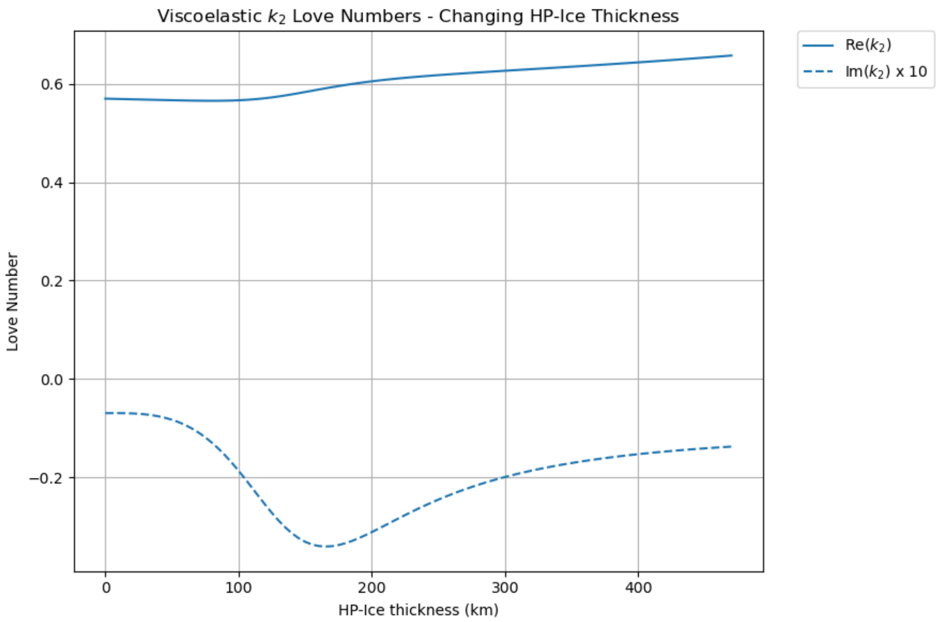

Figure 2: Extracted $\mathrm{Re}(k_2)$ and $\mathrm{Im}(k_2)$ from PyALMA3 for a Titan model with increasing hp-ice thickness from 0km (fully ocean interior) to 475km (fully solid interior) with constant *Ih* shell and core thickness. $\mathrm{Im}(k_2)$ has been multiplied by x10 for easier visualisation with $\mathrm{Re}(k_2)$.

As more rigid material is added at depth the moon becomes stiffer overall, and tidal dissipation increases; liquid layers lower tidal dissipation as they mechanically decouple rigid layers and reduce radial stresses (Tobie *et al.*, 2005), hence $\mathrm{Im}(k_2)$ becoming more negative with increasing hp-ice thickness up to $\tilde{1}70$km. This represents an optimal thickness for tidal dissipation before the ice layer becomes too thick for the forcing timescale, synonymous with the layer becoming dominantly elastic ice rather than viscous liquid and overall reducing the deformability of the body. $\mathrm{Re}(k_2)$ undergoes a similar trend with increasing hp-ice thickness, which can be more clearly seen in Figure 3. The changing Q values with increasing hp-ice thickness can be found in Appendix Figure 7.

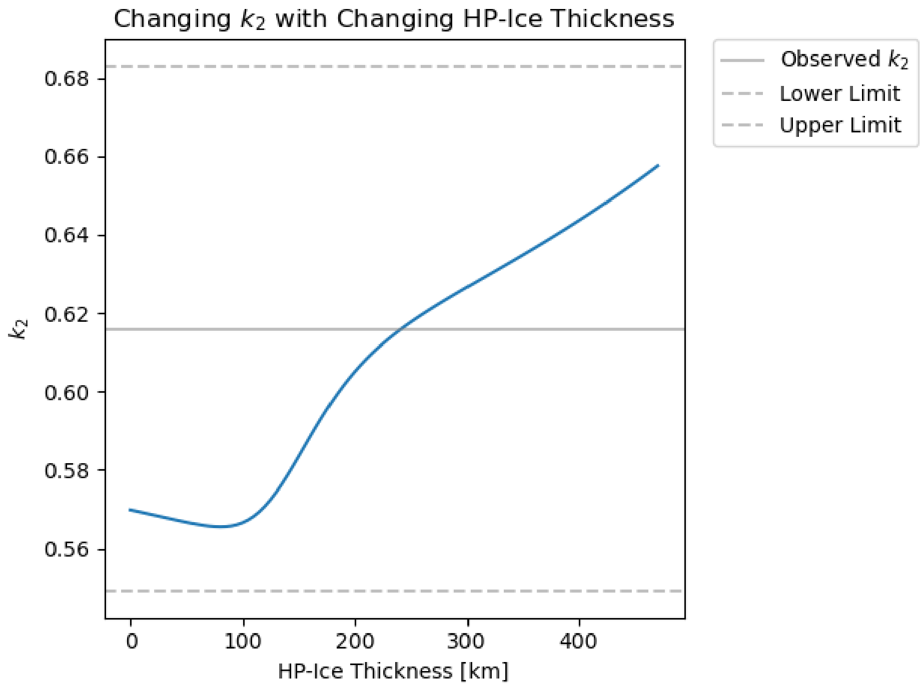

Figure 3: The $\mathrm{Re}(k_2)$ value under increasing hp-ice thickness, with comparison to the published upper limit of observed k_2 from Cassini analysis ($k_2 = 0.616 \pm 0.067$). All iterations of the Titan interior model fit within the uncertainties.

The initial dip in $\mathrm{Re}(k_2)$ around 100km hp-ice thickness is indicative of how introducing a thin ice-layer reduces the overall deformability of the body in response to tidal forcing. However, as ice thickness increases the elastic contribution again dominates the interior and increases the elastic responsiveness of the moon. When comparing the calculated k_2 values with the published k_2 upper and lower limits ($k_2 = 0.616 \pm 067$; $k_2 = 0.375 \pm 0.06$), the PyALMA3 output for models A, B, and C fit into the boundaries of the larger k_2 value rather than the smaller, as shown in Figure 4. A hp-ice thickness of $\sim$300km and ocean layer of $\sim$170km correspond to a k_2 of 0.616, suggesting that the current observed tidal parameters may indeed support the presence of a subsurface ocean, and hence an environment that could support life.

However, considering both the fully-liquid and fully-solid models fit inside the upper k_2 boundary limits, this plot emphasises the challenge in determining interior composition from Love number evaluation. Supporting analysis of the consequent energy dissipation is therefore required to fully evaluate the implications of tidal heating on subsurface habitability.

3.2 Energy Dissipation of Interior Models

The results from the PyALMA3 computation (Section 3.1) were subsequently applied to extract the tidal energy dissipation following the methodology outlined in Section 2.3 A reference logarithmic heat-map (Fig.4) from the published upper and lower limits for k_2 and Q (Table 1) indicated that the largest energy dissipation came from maximum k_2 (0.683) and lowest Q (2.62), as expected from the relationship in Equation 4. Generally, the energy dissipation is in the order of TW, showing consistency with published estimates (Petricca *et al.*, 2025) despite the model being greatly simplified. Figure 4 hence provided the baseline energy dissipation expected at Titan for comparison with the PyALMA3 k_2 and Q values under changing hp-ice thickness.

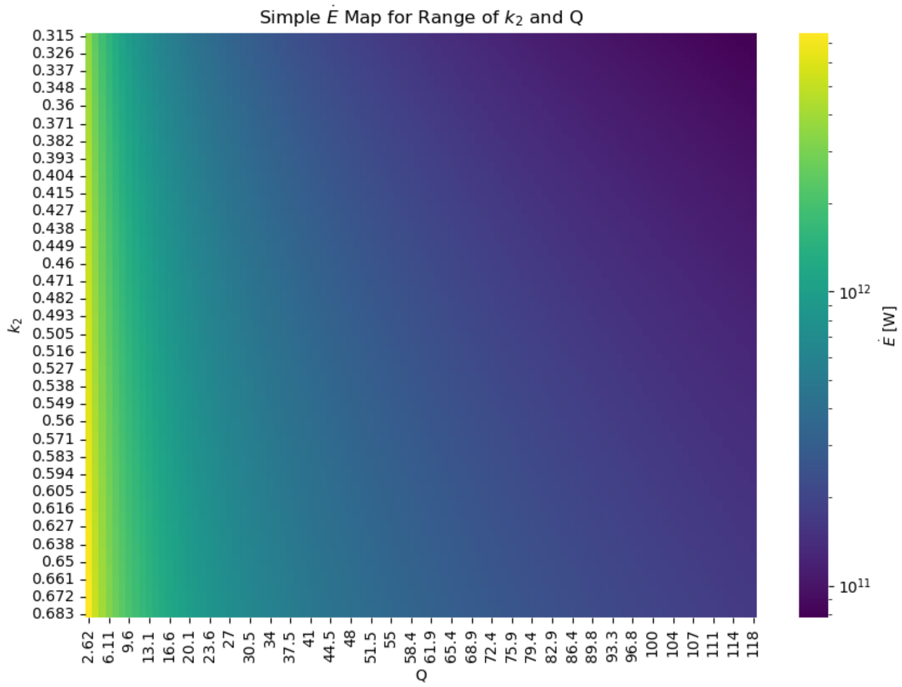

Figure 4: The reference logarithmic heat-map of energy dissipation ($\dot{E}$) generated from the simple $\dot{E}$ equation (Eqn.4), over a range of k_2 (0.315-0.683) and Q (2.62-118) derived from published Titan observations (Table 1). The maximum energy output is in the same order of magnitude (TW) as other estimates for Titan's tidal dissipation (Petricca *et al.*, 2025).

The resulting heat-map for the modelled Titan interiors iterating from fully ocean to fully ice (Fig.5) fit within the set range, therefore aligning the energy dissipation with observational parameters. At lower ice thickness (0km) there is greater Q, indicating a smaller tidal lag and thus lower tidal dissipation; this is likely due to the damping effect of thicker liquid layer on tidal forcing within the interior. This is the argument applied by Petricca *et al.*, 2025 to disprove the presence of a subsurface ocean, suggesting their low value of Q (Q=5) would not be possible with a liquid interior. As the hp-ice shell thickness increases, Q decreases and the rate of energy dissipation increases; this is the same physical mechanism that causes $\mathrm{Im}(k_2)$ to initially become more negative as an ice-layer was introduced (Section 3.1). Energy dissipation increases with increasing ice-thickness until a maximum $\dot{E}$ (9.807×10^{11}W) at 175km, before decreasing in line with the trends observed for $\mathrm{Im}(k_2)$. These results can therefore provide an ideal 4-layer interior (hp-ice = 175km, ocean = 300km) for the greatest tidal heating of Titan, suggesting a high-enough tidal forcing to generate a global melt layer. However, as the hp-ice thickness range (0-475km) within the k_2 and Q limits does allow for a fully solid interior, it remains physically plausible under these model conditions for there to be no subsurface ocean, thus greatly limiting Titan's habitability. Contrastingly, the results also allow for a fully liquid interior, which could arguably increase the habitability potential by facilitating rock-water interactions with Titan's silicate core.

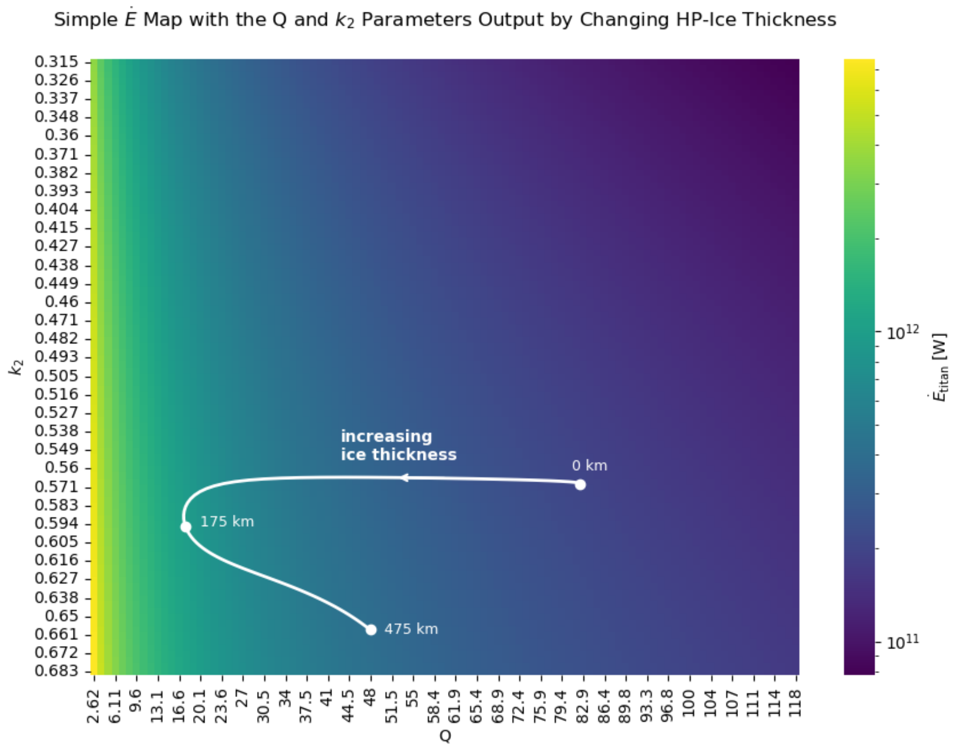

Figure 5: The logarithmic heat-map of energy dissipation ($\dot{E}$) including the extracted k_2 and Q values from the PyALMA3 model. The extracted values fit within the bounds of the current observational limits, with maximum ($\dot{E}$) at an hp-ice thickness of 175km.

3.3 Melting Timescale

Whilst Titan's observed tidal parameters may preclude the availability of a subsurface ocean and facilitate maximum hp-ice shell thickness (Fig.5), it may still be possible for an ocean layer to develop in the future via the heat diffusion from tidal dissipation. The methodology discussed in Section 2.3 was computed to produce Figure 6, extracting the melting timescales for a 475km hp-ice layer with constituent pressure-dependent ice phases (Ice VI, Ice V, Ice III) under Titan's current orbital parameters. The corresponding k_2 and Q values for hp-ice layer thickness of 475km were 0.6575 and 47.91 respectively.

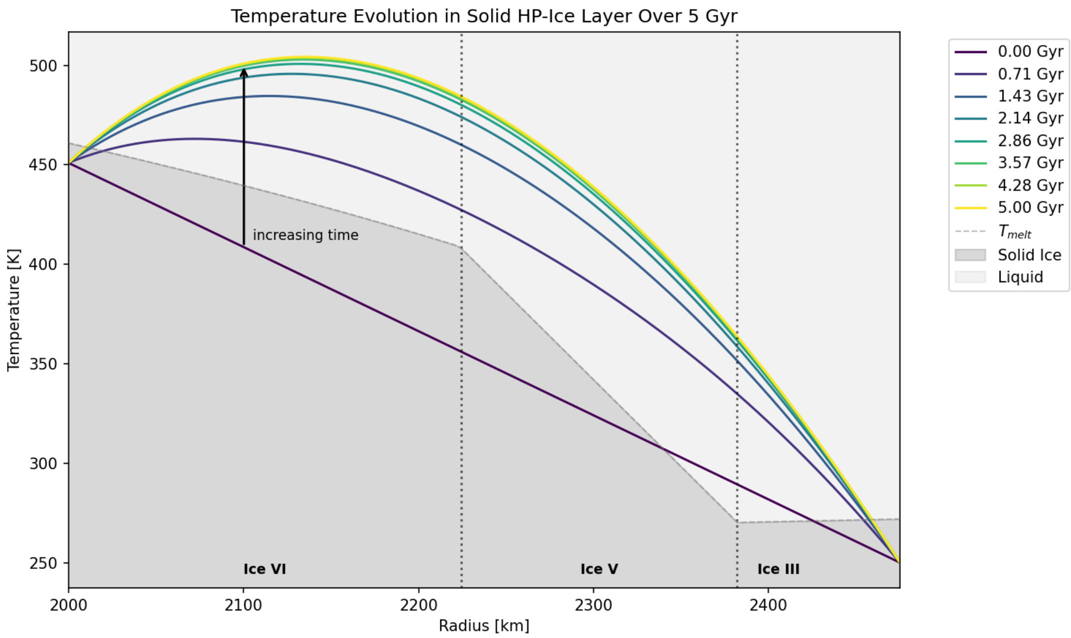

Figure 6: The plot of temperature evolution and subsequent melting for a solid hp-ice interior, with pressure-dependent ice phases. The transition between each ice phase is indicated, with melting temperature generally decreasing with decreasing pressure. A large liquid layer within all ice phases is developed from 0.71 Gyr onwards.

The results indicate that tidal dissipation alone produces enough heat to form a large liquid layer within Titan's interior after 0.71 Gyrs, which is sustained for at least another 4.29 Gyrs until the Sun evolves from the main-sequence. Considering that life on Earth is believed to have taken 300 Myr to develop from Earth's formation (Lahav *et al.*, 2001), the lifetime of this potential subsurface ocean is remarkably significant for establishing habitability on Titan; Titan's interior is arguably stable enough over geological time for prebiotic reactions to occur and for life to be established, if the constituent prebiotic organics are also readily available. As Titan's orbit is changing at a rate of 11cm/yr, the change in energy dissipation with increasing orbital separation over 5 Gyr was also tested to determine if a sustained subsurface ocean was non-viable under such conditions. However, the resulting energy dissipation after 5 Gyr was only 2.66% smaller, and thus had a negligible impact on heat diffusion in the moon's interior.

4 Discussion

4.1 Implications for Habitability

The results of this modelling investigation into Titan's tidal parameters highlighted that Titan is most habitable within the higher observed k_2 regime (Durante *et al.*, 2019), with respect to the availability of a subsurface ocean, with no overlap of the lower k_2 limit (Goossens *et al.*, 2024). Consequently, it can be suggested that future analysis of Cassini gravimetric data or upcoming Dragonfly measurements should aim to confirm that k_2 lies within the upper regime to fully evaluate Titan's interior habitability.

The tidal parameters of k_2 and Q were found to be crucial in determining ocean longevity due to their control over tidal energy dissipation within the interior, hence constraining subsurface melting. Following observational limits for k_2 and Q set by published Cassini analysis, each compositional model of Titan's interior (Model A, B, C) was found to fit within the observational parameter space. This highlights the challenge in determining interior structure and habitability solely from Titan's tidal parameters, as under these model conditions a inhabitable, Model C-like icy interior is just as likely as a fully liquid Model-A interior. A 3-layer, model A structure was suggested by Goossens *et al.*, 2024 as being an optimal interior for promoting habitability due to the contact between the silicate core and ocean facilitating rock-water interactions associated with origin-of-life processes on Earth (Vance and Melwani Daswani, 2020). Consequently, it can be argued that the current limits of Q allow for the possibility of such interactions occurring on Titan and therefore increasing the likelihood of prebiotic reactions occurring. However, this is dependent on there being no hp-ice layer present, which goes against the current consensus of Titan's interior structure (Petricca *et al.*, 2025). Regardless, as the identification of a stable subsurface ocean was the main objective of this investigation to determine habitability, the resulting k_2 and Q range still facilitate this.

The modelling of energy dissipation over time highlighted the possibility of Titan developing a stable subsurface ocean within 0.71 Gyr and sustaining it for at least 4.29 Gyrs, regardless of orbital separation changes. This is highly significant for establishing global habitability on the moon, as a sustained liquid layer with a constant energy source and organic constituents provides an ideal environment for prebiotic reactions. As a result, it can be implied that a present-day, solid-ice Titan may still develop an interior ocean within its current orbital lifetime and provide a key environment for supporting life in the Solar System sometime in the future.

4.2 Limitations

4.2.1 Observational Constraints

The main limitations of this investigation come from the Titan's current observational constraints and uncertainties of its orbital history. At present, thermal evolution models are the only prediction of Titan's differentiated interior (Tobie *et al.*, 2005), resulting in multiple assumptions being made that are unlikely to be resolved before the upcoming Dragonfly mission arrives. Additionally, the ongoing debate about Cassini's k_2 measurements (Durante *et al.*, 2019; Goossens *et al.*, 2024) and Q being poorly constrained (Downey and Nimmo, 2025) introduces large uncertainties when considering the validity of the model predictions. The minimum ice-thickness set by the boundary condition of Q in Figure 5 may therefore not truly reflect Titan's maximum Q value, consequently altering assumptions made about the requirement of a hp-ice layer in the interior.

4.2.2 Unknown Interior Structure

Whilst the composition of the core and ice layers within Titan are generally agreed (Petricca *et al.*, 2025), there is ongoing debate about the density of any existing ocean layer (Goossens *et al.*, 2024). This study assumed a high-density ocean layer for iterations from Model A to C, however using a more water or ammonia-rich composition would have resulted in a lower Love number from PyALMA3, hence altering the subsequent energy dissipation calculations. The extracted k_2 and Q values are inherently reliant on assumed interior parameters, and can produce similar outputs for various compositions, presenting a degeneracy in determining interior structure from tidal parameters alone.

For simplicity, this investigation assumed layers of uniform density and composition with a 1D spherical symmetry, therefore neglecting the influence of lateral variations on tidal response and ignoring alternate methods of heat transfer like convection. When calculating the tidal parameters from PyALMA3, the hp-ice layer was assumed to have a constant density regardless of ice phase, which would otherwise vary with pressure and temperature. The boundary temperatures between the core→hp-ice layer and the hp-ice layer→*Ih* shell were also fixed and did not account for the temperature evolution of the core or ice shell under tidal dissipation. The rheology of the interior was also simplified and assumed to be constant throughout the respective layers, which may not fully capture the behaviour of these materials under these pressure and temperature conditions.

4.2.3 Tidal Heating Changes

As k_2 and Q are dependent on interior structure, they are not constant with time as the interior evolves. This model has however assumed constant tidal parameters, and a corresponding uniform tidal dissipation throughout the hp-ice layer. Whilst uniform heating has been applied to other thermal evolution studies, this still neglects the reality of heterogeneous energy distribution within the interior (Tobie *et al.*, 2005) and it cannot be assumed that Titan's tidal dissipation is even without further observations. Additionally, no changing orbital parameters were considered apart from Titan's increasing orbital separation (Lainey *et al.*, 2020), and this was also taken to be constant over time. Titan's current orbit is believed to have evolved from an excitation event ∼350 Myrs ago, with its eccentricity and inclination being rapidly damped (Downey and Nimmo, 2025) however this remains challenging to constrain. As eccentricity exerts a large control over tidal dissipation, accounting for its decrease from tidal effects would provide a more comprehensive view of Titan's internal temperature evolution over time.

5 Conclusion

Overall, Titan's tidal parameters k_2 and Q exert great control over both the detectability of a subsurface ocean and energy dissipation from tidal forcing. Their measurement is therefore directly tied to determining Titan's habitability, which is largely reliant on the availability of a liquid interior layer to host an environment for prebiotic reactions and an interface for surface-interior cycling. Despite ongoing observational constraints and debate over the true value of k_2 from archival Cassini data, the output Love numbers for various model interiors aligned solely with the higher published k_2 value 0.616 ± 0.67. Future Titan habitability studies should therefore seek to use this, more widely accepted, k_2 value as a best fit for body deformability rather than the recent suggested value of 0.375 ± 0.06; whilst it is possible for Titan to be inhabitable, solid-ice under the higher k_2 parameter, it also has the potential to support a liquid layer, which the lower k_2 did not provide any evidence for under these modelling conditions.

Whilst Titan's current tidal parameters may preclude the presence of a subsurface ocean, their influence on interior melting timescales allows for the possibility of a more habitable Titan emerging within the main-sequence lifetime of the Sun. The influence of tidal heating on Titan's interior melting potential is therefore greatly significant for establishing a present, or future, global habitable environment able to sustain and support origin-of-life interactions. It is important to note that the assumptions applied for Titan's interior structure, in both modelling the tidal parameters and subsequent energy dissipation, are limiting and may not reflect Titan's true physical structure. However, the results of this investigation have still succeeded in validating appropriate tidal parameters for determining habitability and subsurface ocean availability. Future work should aim to consider alternate methods of heat transport within the interior, lateral and radial layer variations, and further orbital changes to comprehensively characterise Titan's tidal dissipation and melting timescales.

Despite the uncertainties surrounding Titan's true interior composition, which are unlikely to be resolved before the upcoming Dragonfly mission, the possibility of Titan hosting a global subsurface ocean remains an exciting area of investigation for habitability in the Solar System.

6 Appendix

Table 2: Physical constants and model parameters used to model Titan's interior and tidal heating. Unless otherwise noted, values are taken from Petricca *et al.*, 2025, Supplementary Information S11.

Symbol	Description	Value	Source / Notes
Bulk properties			
R_{Titan}	Titan radius	2575 km	
ρ_{ice}	Ice crust density	920 kg m^{-3}	
ρ_{ocean}	Ocean density	1200 kg m^{-3}	
$\rho_{\text{hp-ice}}$	HP-ice layer density	1400 kg m^{-3}	
ρ_{core}	Core density	2600 kg m^{-3}	
ρ_{avg}	Bulk (average) density	1882 kg m^{-3}	
g_{Titan}	Surface gravity	1.35 m s^{-2}	
ω	Orbital frequency	4.56×10^{-6} rad s^{-1}	
M_{Titan}	Mass of Titan	$\rho_{\text{avg}} \cdot \frac{4}{3}\pi R_{\text{Titan}}^3$	Derived
Layer structure			
R_{core}	Rocky core radius	2000 km	Elastic rheology
R_{shell}	Icy shell thickness	100 km	Maxwell rheology
R_{ocean}	Ocean layer thickness	$0 - 475$ km	Newtonian rheology
$R_{\text{hp-ice}}$	HP-ice layer thickness	$0 - 475$ km	Maxwell rheology
Rigidity (shear modulus)			
μ_{ice}	Ice shell rigidity	3 GPa	Julia Code
μ_{core}	Core rigidity	30 GPa	Julia Code
$\mu_{\text{hp-ice}}$	HP-ice layer rigidity	6 GPa	Julia Code
Viscosity			
η_w	Ocean viscosity	1×10^4 Pa s	Near-freezing / brine
η_m	Basal ice viscosity	1×10^{13} Pa s	
η_{hp}	HP-ice viscosity	1×10^{11} Pa s	
Thermal parameters			
T_m	Basal ice-shell temperature	250 K	Ice *Ih* melting point
T_s	Surface temperature	94 K	Jennings *et al.*, 2019
E_a	Activation energy (ice VI)	61.9 kJ mol^{-1}	Noguchi and Okuchi, 2020
R_g	Universal gas constant	8.314 J mol^{-1} K^{-1}	
Physical / orbital constants			
G	Gravitational constant	6.6743×10^{-11} m^3 kg^{-1} s^{-2}	
σ	Stefan–Boltzmann constant	5.6703×10^{-8} W m^{-2} K^{-4}	
M_s	Mass of Saturn	5.683×10^{26} kg	
e	Eccentricity of Titan	0.0288	
a_t	Semi-major axis of Titan	$1,221,870$ km	
n	Orbital mean motion (rotation rate)	$\sqrt{GM_s/a_t^3}$	Derived

Table 3: Additional physical constants and thermal/pressure parameters used in the HP-ice layer melting and temperature evolution model.

Symbol	Description	Value	Source / Notes
Pressure calculations			
P_{surface}	Titan atmospheric surface pressure	147 kPa	Harri *et al.*, 2006
$P_{\text{base, shell}}$	Pressure at base of ice shell / top of HP-ice layer (no ocean)	0.898 GPa	Derived to be within the ice VI stability field (0.6–2.2 GPa)
HP-ice thermal properties			
k_{cond}	Thermal conductivity	2.5 W m^{-1} K^{-1}	
c_p	Specific heat capacity	2100 J kg^{-1} K^{-1}	
α	Thermal diffusivity	$\frac{k_{\text{cond}}}{\rho_{\text{hp-ice}}\, c_p}$	Derived
HP-ice layer structure and boundary conditions			
$R_{\text{hp-ice}}$	Outer radius of HP-ice layer	475 km	
R_{inner}	Inner (core) boundary radius	2000 km	Fixed
T_{surface}	Surface temperature	94 K	Fixed, Jennings *et al.*, 2019
T_{outer}	Temperature at base of ice shell	250 K	Fixed, melting point of Ice *Ih*
T_{inner}	Temperature at core boundary	450K	Kept just below local ice VI melting point (461K)[1]
Ice Ih–VI–VII melting curve (Simon-Glatzel equation) parameters			
T_0	Reference temperature	273.31 K	Choukroun and Grasset, 2010
P_0	Reference pressure	0.000612 GPa	Choukroun and Grasset, 2010
a	Simon equation constant	0.1835 GPa	Choukroun and Grasset, 2010
c	Simon equation exponent constant	3.6	Choukroun and Grasset, 2010
Melting-temperature curve boundaries			
–	Ice Ih linear slope (valid $P < 0.210$ GPa)	-0.0098 K MPa^{-1}	From $T_m = 273.16K$
–	Ice Ih - III transition region	0.210–0.3 GPa	Linear interpolation
–	Ice III - V transition region	0.3 – 0.6	
–	Ice VI stability field (Simon-Glatzel equation)	0.6–2.2 GPa	

1. The estimated core temperature for Titan is 500-800K (Castillo-Rogez and Lunine, 2010, yet as this is above the melting point for a hp-ice layer at this depth a lower temperature was considered for the boundary. 450K is therefore a colder estimate of Titan's core-boundary.

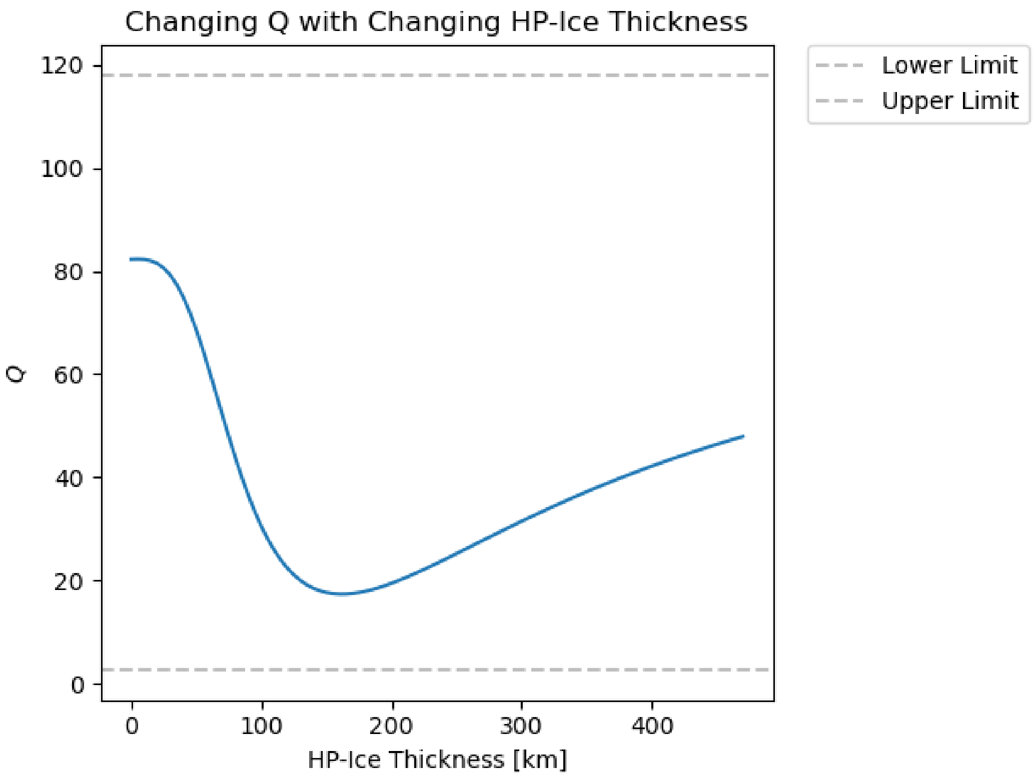

Figure 7: The changing Q value with increasing ice-shell thickness. The trend is similar to the one observed for Im(k_2), and is explored more in depth in Section 3.2 with regards to energy dissipation.

The below Figures are examples of output from the adapted Julia code, exploring changing surface thickness under. While not relevant to the final interior models examined in this investigation, they provided a reference point for later PyALMA3 use.

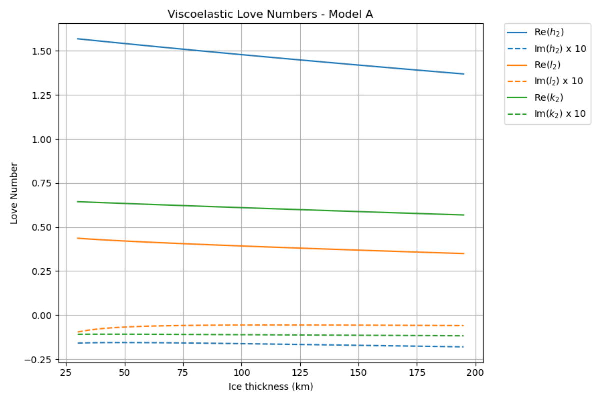

Figure 8: The PyALMA3 output of a 4 layer interior with a thin subsurface ocean (5km).

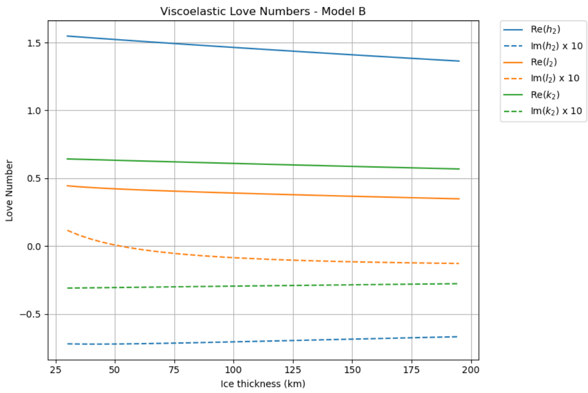

Figure 9: The PyALMA3 output of a 3 layer interior with a thick hp-ice layer (475km).

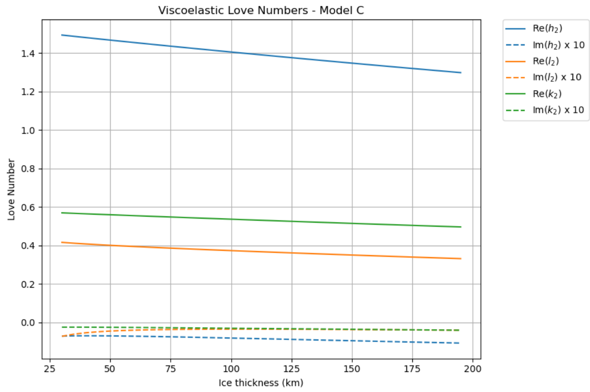

Figure 10: The PyALMA3 output of a 3 layer interior with a thick ocean layer (475km).

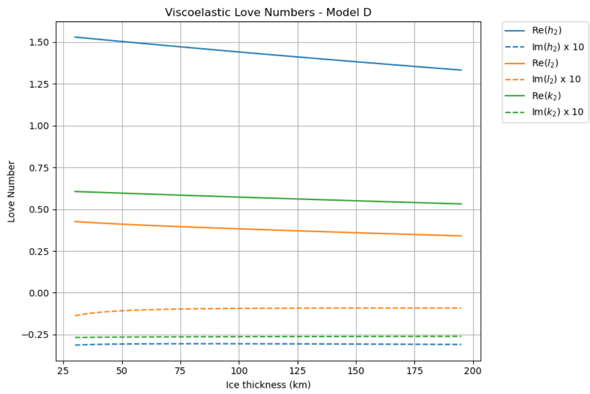

Figure 11: The PyALMA3 output of a 4 layer interior with equal ocean and hp-ice layers.

7 Software

The PyALMA3 (Python plAnetary Love nuMbers cALculator) used in this investigation can be accessed at: https://github.com/drsaikirant88/PyALMA3/tree/main.

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